

Investigate the impact of Confounding Medical Interventions (CMI) in electronic health records (EHR) data

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Background & Motivation

Electronic health records (EHRs) data are widely used for clinical research. The abundance of available data, the low cost of obtaining those observational data, and its real-time nature have given EHR enormous potential to develop powerful predictive models. Researchers rely on EHR data extensively to predict the risk of clinical outcomes. As clinical prediction models (CPMs) find their way into clinical practice, it is important to develop systems and procedures to monitor their safety and effectiveness. The Food and Drug Administration (FDA) has developed draft guidance for how such tools should be considered. Within Duke University Health System, a governance framework has been developed for evaluating and monitoring the performance of CPMs to ensure their adequate performance.

After implementing a CPM, medical professionals will use these models to inform clinical decision-making and allocation of interventions. Ideally, the intervention will prevent the outcome that the CPM is predicting. This leads to an important challenge: how do we evaluate CPMs in the presence of interventions? If the CPM is accurate, and the intervention is effective, those who are truly at risk of the adverse outcome will not actually have the event. To evaluate the CPM appropriately, we are interested in what the causal inference refers to as the counter-factual: the outcome that would have occurred without the intervention being deployed. Unfortunately we do not observe this counter-factual.

Current research on EHR data has investigated several sources of biases such as selection bias informed presence bias, and confounding, and proposed approaches to control these biases. However, little work has considered this potential bias which has been labeled: Confounding by Medical Interventions (CMIs). The little literature on this topic has focused on how we can learn new CPMs in the presence of CMI. The suggestions to mitigate this bias include using data with no CMI for model training or treating all observations with CMI as unhealthy (based on the trust in medical professional's judgement). Although the researchers were able to improve the classification accuracy using these methods, ignoring data with CMI during training reduced the generalizability of the model to validation data, and assigning all data with CMI with negative labels did not perform well enough. As such, the literature is not well developed. This project seeks to study how CMI can impact our ability to evaluate CPMs and strategies we can use to address these challenges. We first propose possible causal models. Next, we use simulation to both illustrate the problem and explore potential solutions. Then, we will evaluate these approaches in real-world healthcare data.

Exploratory analysis

During the initial stage of investigation, we proposed 2 possible causal models.

The basic variables in the causal models are initial risk, intervention (1 = intervention, 0 = no intervention), prevalence of disease, and observed outcome (1 = sick, 0 = healthy). In a simulated scenario, a patient with initial risk higher than the intervention cutoff (CMI) will receive intervention treatment. The effectiveness of treatment is decided by $b_{intervention}$. In general, a higher $b_{intervention}$ means lower effectiveness. We simulate three levels of disease prevalence rate: high (~50% population have the illness without intervention), moderate (~25% population have the illness without intervention), and low (~4% population have the illness without

intervention). A highly prevalent disease includes hypertension, a moderately prevalent disease includes Hepatitis B, and a low-prevalent disease includes $\square$. What differentiates the 2 models is the mechanism to generate the final observed outcome with intervention. The first model calculates a new risk (risk after intervention) for patients who should have had outcome due to higher initial risk and also received intervention treatment. According to the new risk, the final observed outcome is generated probabilistically. The second model doesn't consider risk after intervention. Instead, depending on the intervention effectiveness factor $b_{intervention}$, $(1-b_{intervention})$ % of the population will be assigned to have no outcome (healthy), which means their intervention is successful.

In the real world, only the initial risk and observed outcome are known to us. With the confounding intervention variable, we ideally want to achieve a prediction power as good as predicting outcome directly from the known initial risk. The metrics (AUC, MSE, calibration curve slope) to evaluate CPM are averaged over 100 simulations.

Causal diagram 1

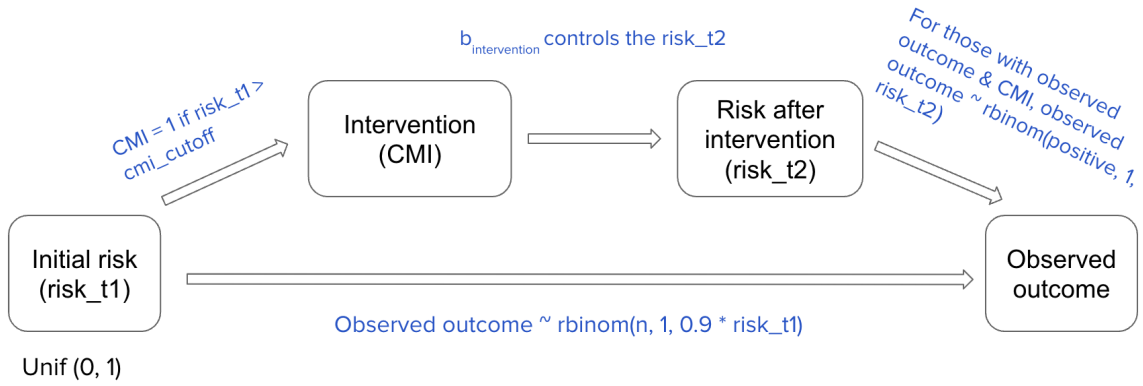
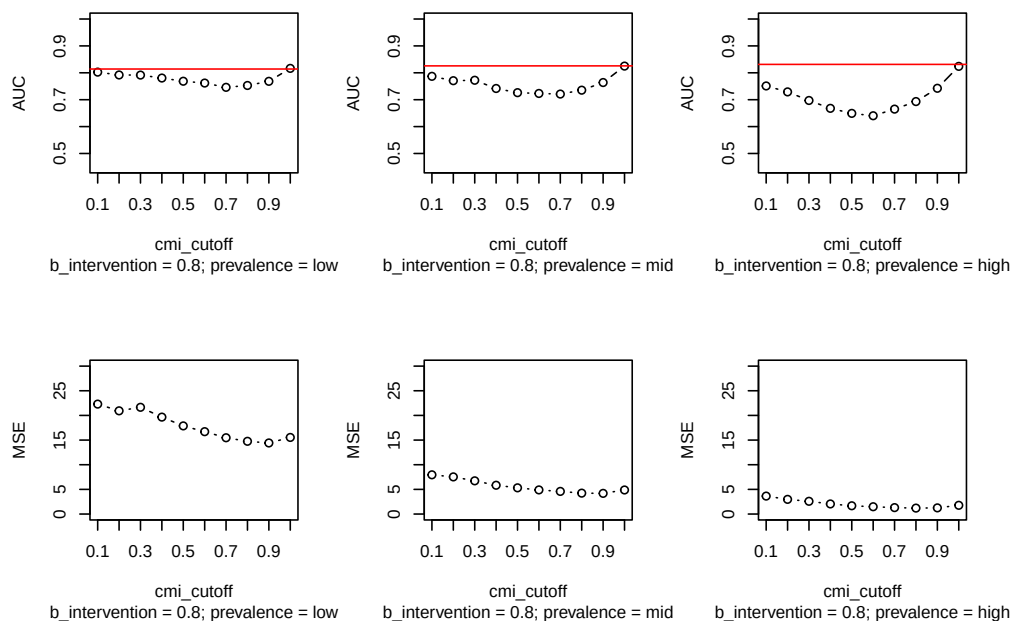


Figure 1: Causal diagram 1

Observed outcome is determined by a binomial distribution as following:

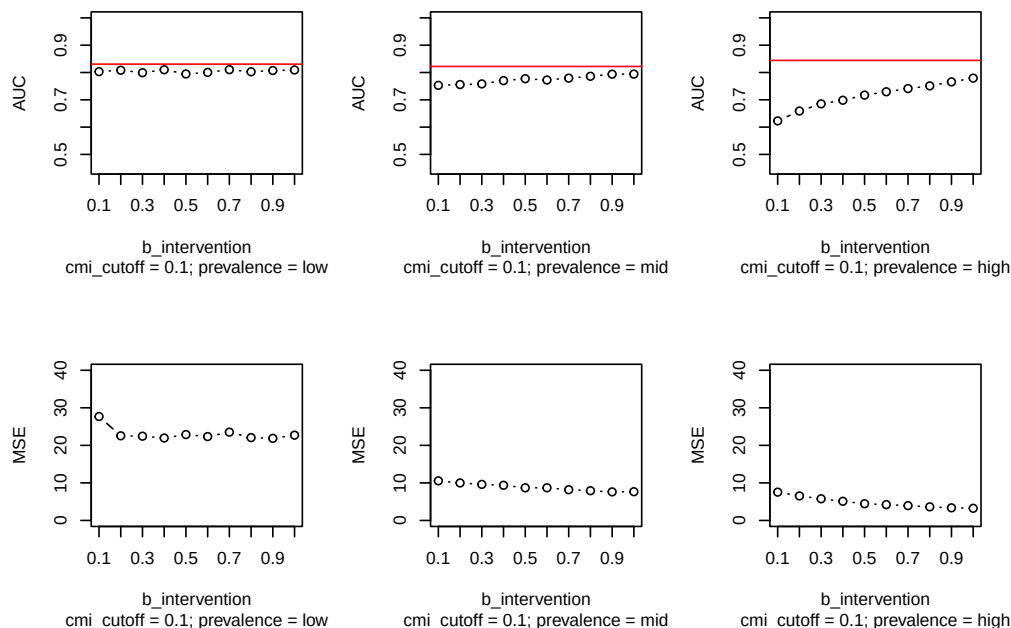
$x = prevalence + 5 * risk_{t1}$ where 5 is a constant to control for $AUC = 0.8$ $p = exp(x)/(1 + exp(x))$
 $outcome = binomial(n, p)$

Vary cmi_cutoff (criteria of giving out treatment)



Note that the higher the cutoff, less people are getting the intervention. In general, when very few or the majority of the population receives treatment, the prediction power is the best. As a disease becomes less prevalent, the overall prediction accuracy and MSE depends less on the percentage of people receiving treatment.

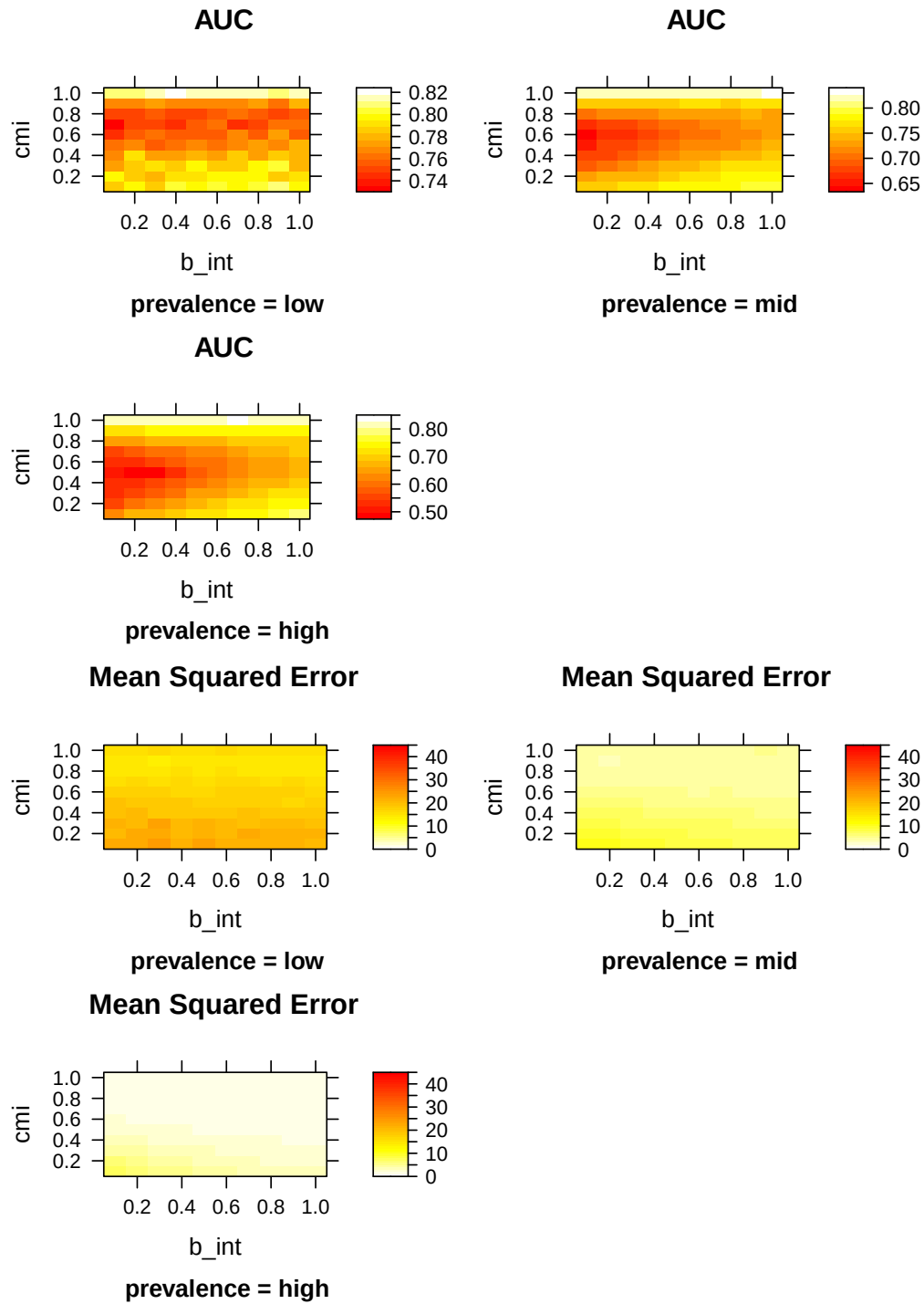
Vary b_intervention (effectiveness of treatment)

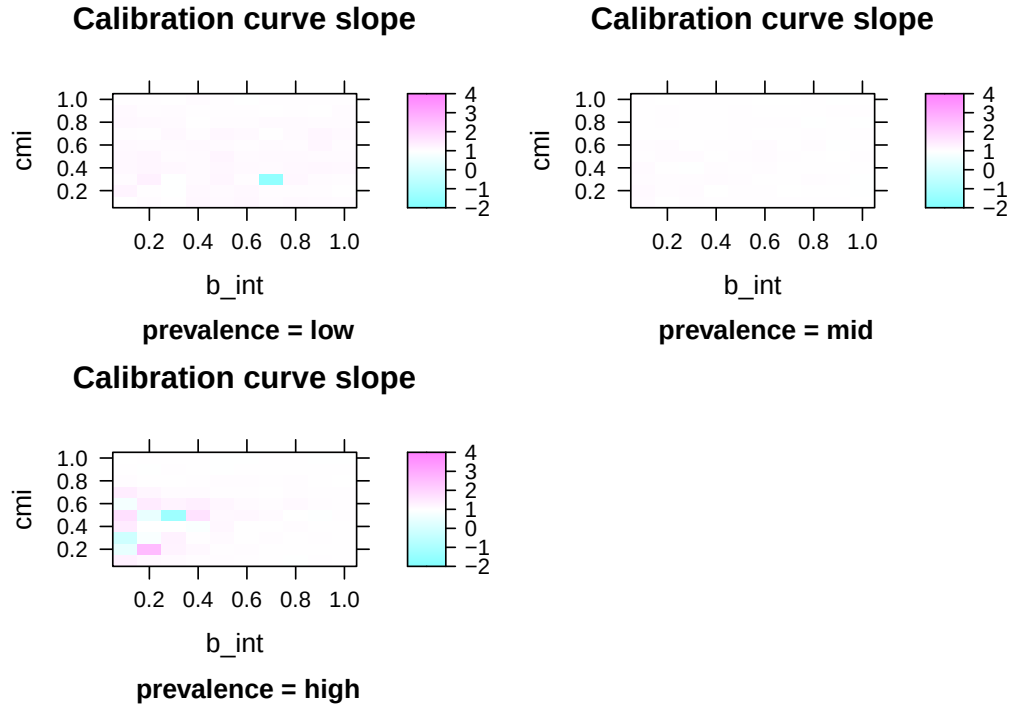


The red line in the top panels denotes the ideal prediction power of fitting the observed outcome directly on the initial risk. In the bottom panels, the mean squared error (MSE) is calculated as the difference between the actual outcome and $\text{logit}(\hat{p})$ (the linear combination $x = \beta_0 + \beta_1 * \text{risk}_{t1}$). As the intervention effectiveness decreases, prediction power increases and MSE decreases. A strong treatment is associated with relatively poor prediction on a patient's final outcome.

As a disease becomes less prevalent, the overall prediction accuracy and MSE depends less on the effectiveness of treatment. The higher the prevalence rate, delivering effective treatment makes the model perform worse.

Varying b_int and cmi_cutoff





Moving from bottom to top, if close to half of population receives treatment, the prediction performance is the worst. However, if almost everyone or no one receives treatment, we see better performance. Moving from left to right, as intervention effectiveness decreases, AUC increases and MSE decreases. However, these trends are only clear in mid to high prevalent diseases. As prevalence rate increase, MSE decreases overall.

Calibration curves evaluate how calibrated a classifier is. From the slope of the curve, we can see whether there is any trend of overestimation or underestimation. The goal is to plot the actual outcome of test data against the linear predictor $\beta_0 + \beta_1 * y.pred$. $y.pred$ is the outcome predicted from test data using logistic regression model trained on $risk_t1$ using the training data. If a model is well calibrated, the slope $\beta_1 = 1$. $\beta_1 > 1$ means underestimation, and $\beta_1 < 1$ means overestimation.

For low to mid prevalent diseases, there is an overall overestimation. For high prevalent disease, there are signs of overestimating the probability of outcome when a treatment is more effective; however, when treatment is less effective, the model becomes better calibrated.

Causal diagram 2

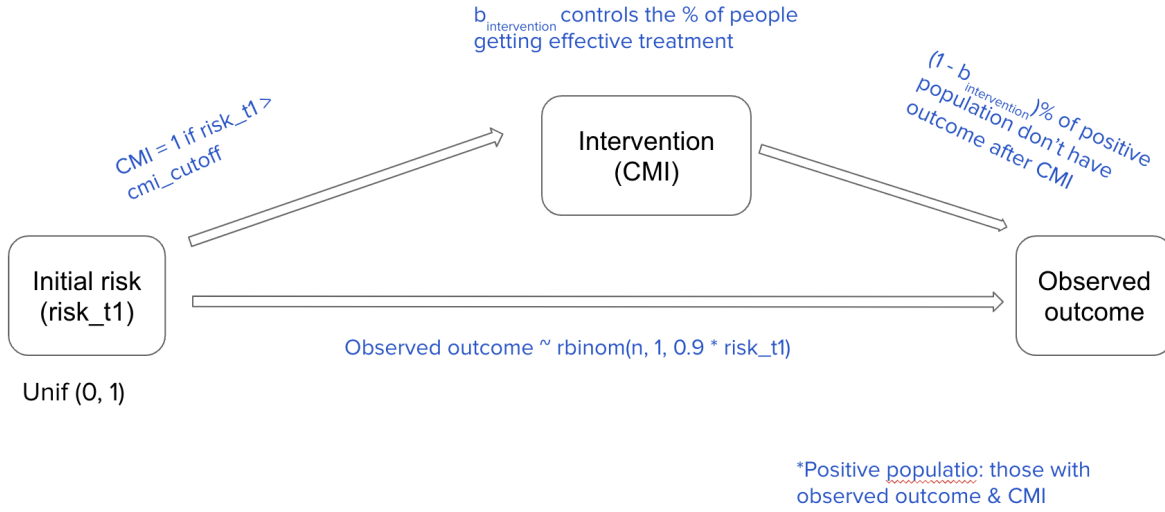
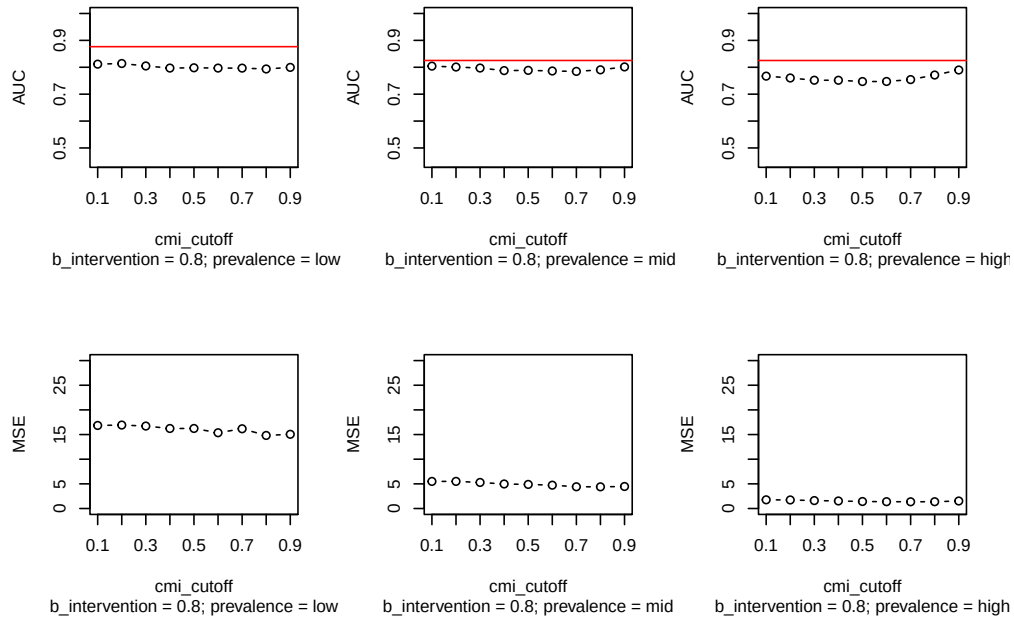


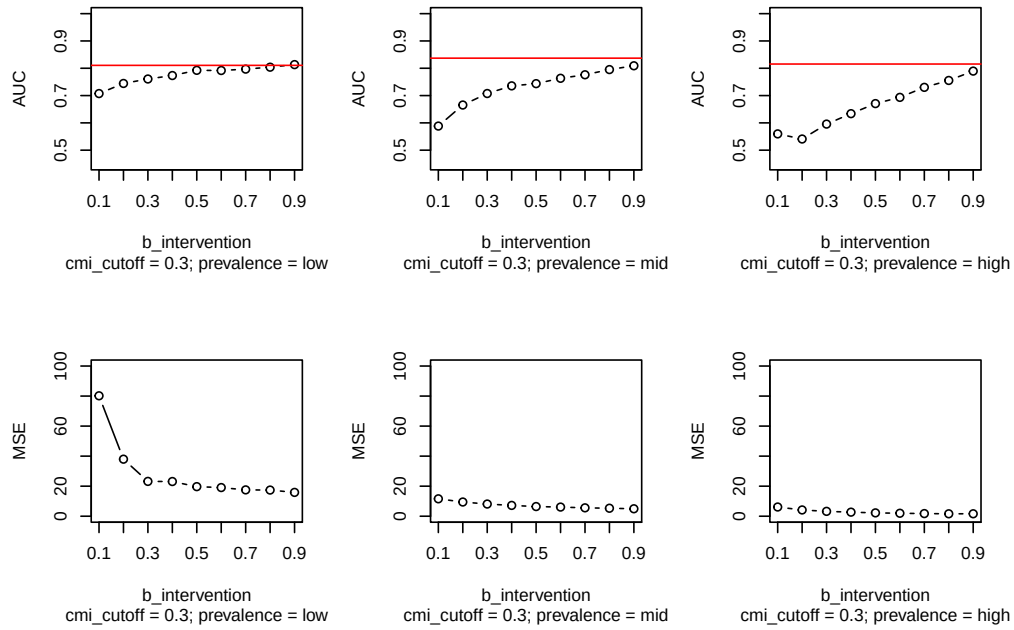
Figure 2: Causal diagram 2

vary cmi_cutoff

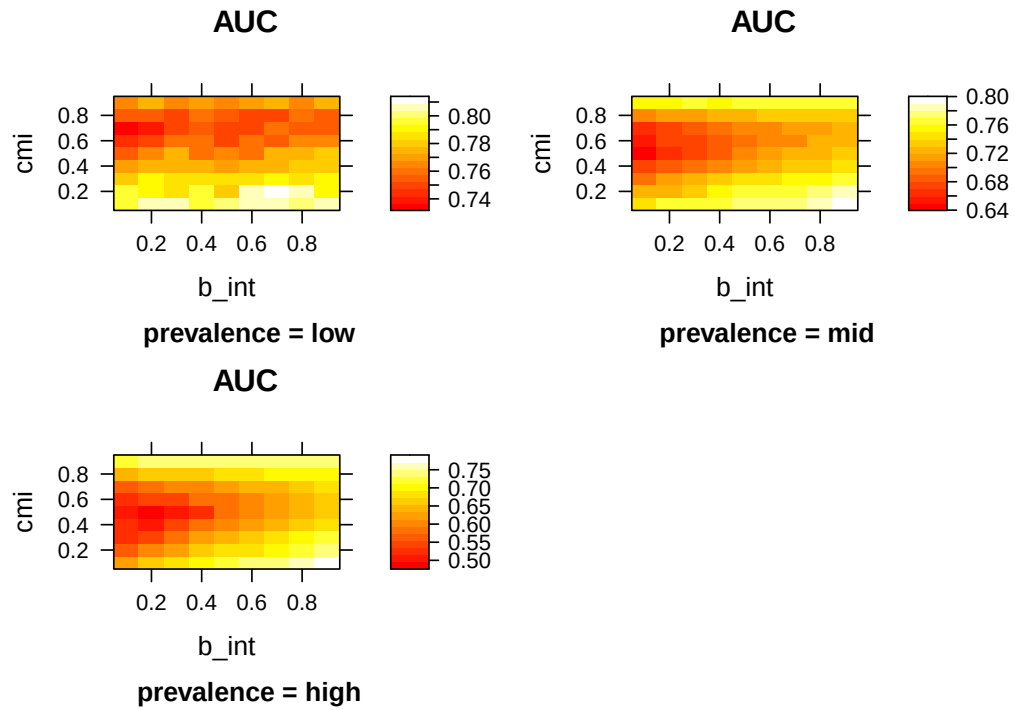


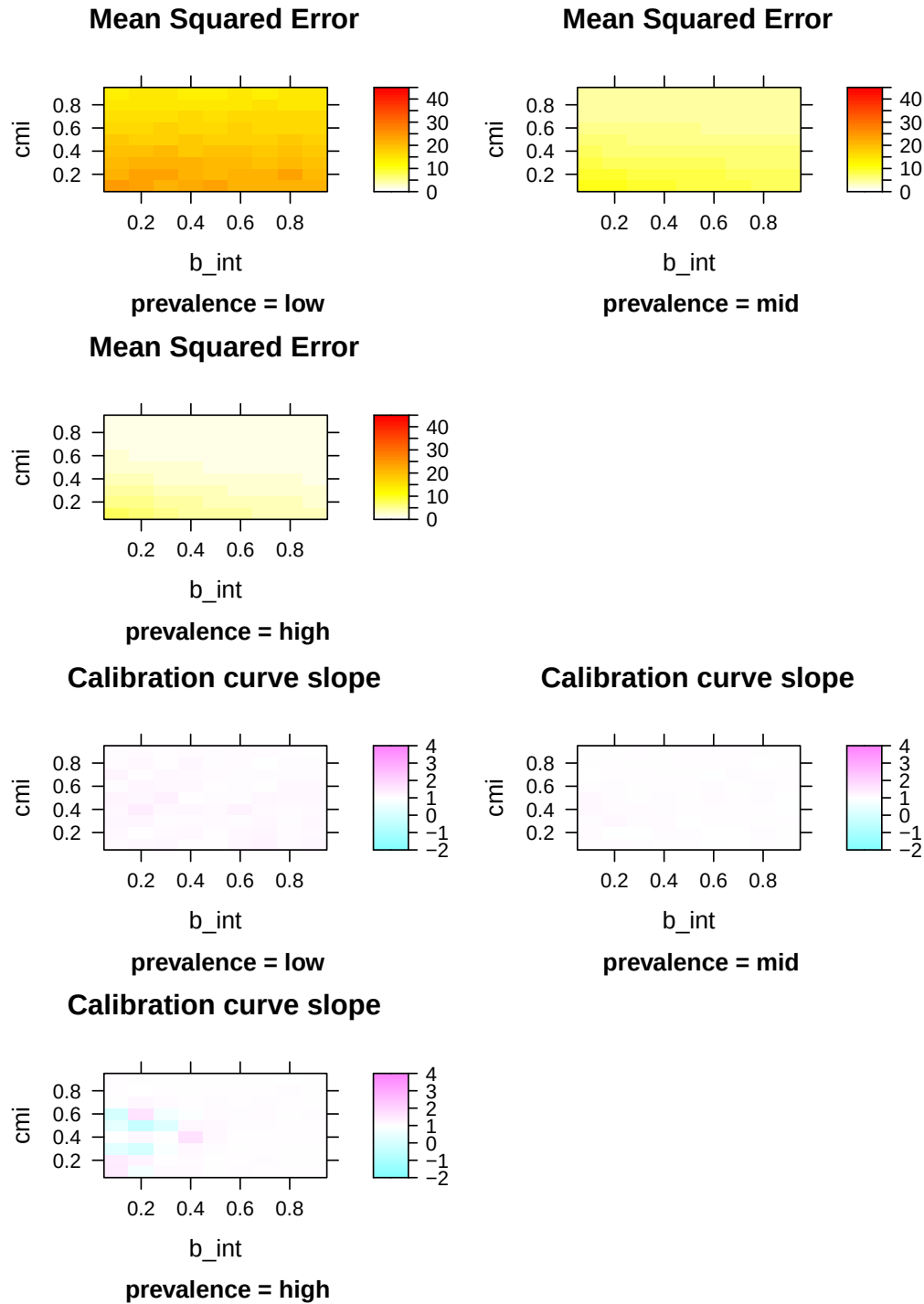
MSE and AUC shows similar trend as the first causal diagram.

vary b_intervention



Vary both b_intervention and cmi_cutoff





Trends are similar to the first causal model.

Important variables

Research question

From the exploratory analysis, we found that as the effectiveness of intervention treatment increase or the prevalence of disease decrease, the predictive performance of our glm model has decreased and MSE has increased. When less people receives treatment, there is less error in prediction. These discoveries circle

back to our research question — evaluating CPMs in the presence of intervention. CPM prefers (produce better predictive performance and lower error) high-prevalence disease, poor treatment, and small coverage of intervention.

What to do next - with the goal being improving predictive power * explore other classification model other than glm, such as random forest or boosting * As treatment become stronger, there end up to be fewer positive samples. Maybe we can try upweighing positive samples as we increase the power of treatment * decompose total effect into direct and indirect effects * <https://cran.r-project.org/web/packages/foreach/vignettes/foreach.html> optimize time complexity

foreach library to optimize time complexity